EnergyCopilot: A Context-Aware, Bidirectional LLM Assistant for Occupant Comfort, Health and Building Energy Efficiency

Concept note

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1. Problem

Buildings use about 30% of global final energy [1], and people spend nearly 90% of their time indoors [2], where temperature, humidity, CO_2 and PM2.5 affect comfort, health and cognitive performance [3]. Occupant behaviour drives a large share of both energy use and indoor conditions, and timely, explained feedback can save energy [4]. Yet the tools available to occupants fall short: BMS dashboards show numbers without explaining them, and general-purpose LLM chatbots speak fluently but know nothing about *this* building and may hallucinate [5]. Recent work covers pieces of the problem, such as LLM Text-to-SQL over HVAC data [6], question answering over Brick graphs [7], and Graph-RAG over energy-efficiency guidance [8], but no system brings them together.

2. Core idea

EnergyCopilot is a conversational assistant that *knows the building* and *talks in both directions*:

- **Pull (answer):** occupants and facility managers ask in natural language, e.g., *"Why is room B203 so stuffy this afternoon?"*, and receive an answer backed by the room's own sensor data, the relevant standard and a concrete action.
- **Push (advise):** the assistant watches sensor streams, detects waste or discomfort (e.g., air-conditioning on with the window open), and sends a short, explained nudge with the expected saving, then learns from whether the user accepts it.

The guiding principle is **"tools first, language second"**: the LLM plans and explains, while every number comes from a deterministic, auditable tool (Figure 1):

(1) **Text-to-SQL over sensor data**, with rooms and points resolved through a Brick semantic model [9];

(2) **Graph-RAG over expert knowledge**, covering standards (ISO 7730, ASHRAE 55, WHO, energy codes), textbooks and expert rules, with citations;

(3) **Comfort and health assessment**, i.e., PMV/PPD, the adaptive model and IAQ thresholds;

(4) **What-if simulation** (EnergyPlus or a fast surrogate), which checks each candidate action a before it is recommended:

$$a^* = \arg\min_{a \in \mathcal{A}} \left[\Delta E(a) - \lambda \log \hat{p}_u(a)\right] \;\text{ s.t. }\; |\mathrm{PMV}_z(a,t)| \le 0.5, \;\; C_z^{\mathrm{CO_2}}(a,t) \le C_{\max}, \tag{1}$$

where ΔE is the predicted energy change and $\hat{p}_u$ is the learned probability that user u accepts the action;

(5) **Forecasting and anomaly detection**, which trigger the proactive loop.

A verifier re-checks queries, units and citations before any answer is shown.

3. What is new

- **One loop for data, semantics and knowledge.** Sensor data, the building's semantic model and expert literature are combined in a single conversation.
- **Impact-quantified advice.** Recommendations are simulated and comfort-constrained, not just generated.
- **Bidirectional and personalised.** The assistant answers when asked, speaks up when needed, and adapts to each user.
- **Open benchmark.** *EnergyQA*, a multi-task benchmark covering data queries, comfort and health knowledge, diagnosis and recommendation, to make such assistants comparable.

4. Expected impact

The project aims to deliver energy savings without loss of comfort, healthier indoor air through timely, understandable alerts, and occupants who understand their building. It will produce a validated prototype, open datasets and code, and evidence from a controlled field pilot.

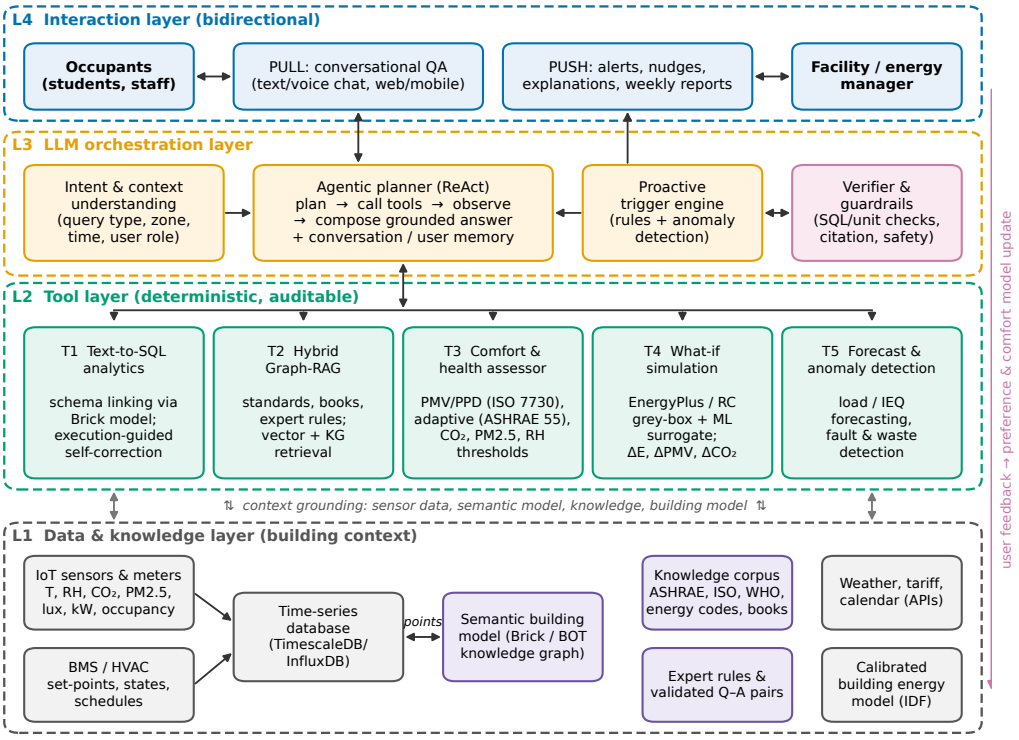


Figure 1. EnergyCopilot concept. The LLM (L3) plans and explains; deterministic tools (L2) compute and retrieve from the building context (L1); users interact in both directions (L4) and their feedback personalises future advice.

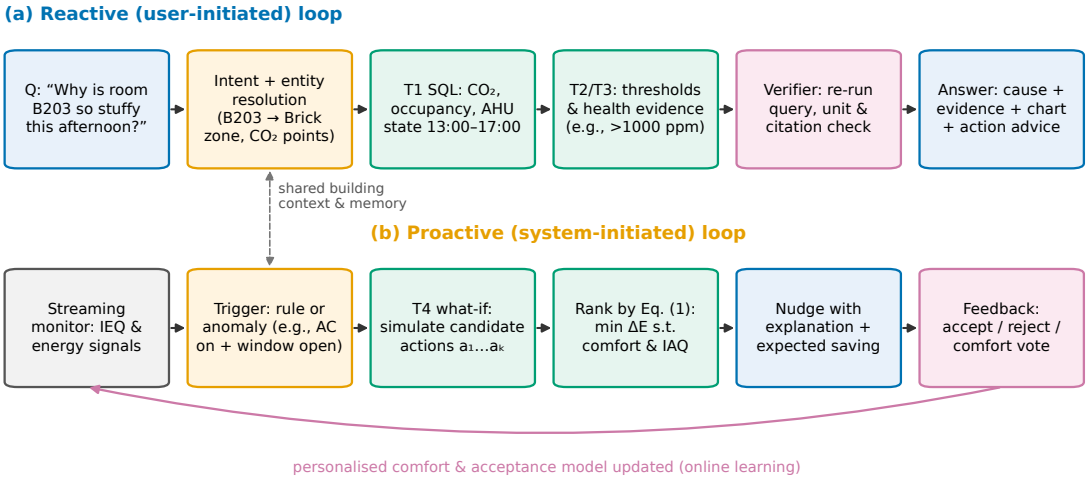


Figure 2. (a) Reactive loop: the user asks and the assistant answers with evidence. (b) Proactive loop: the assistant detects an event, simulates candidate actions, ranks them with Eq. (1), nudges, and learns from feedback.

5. Open questions for domain experts

The copilot is only as good as the domain knowledge behind it. A preliminary search (Table 1) found useful *evidence* sources, but **no open, occupant-facing question–answer dataset** on how indoor conditions affect comfort, health and performance, or on everyday energy-saving behaviour. Existing Q&A benchmarks target building metadata [7], AEC engineering knowledge [10] or the energy sector broadly [11]. We would value expert input on the following questions.

E1. Q&A datasets. Do you know of datasets, FAQ collections or curated knowledge bases, public or institutional, with questions and answers on occupant-facing topics? Examples include the positive and negative effects of high or low temperature and humidity, how harmful high CO_2 is and at what level, PM2.5 exposure, daylight and glare, and practical energy-saving actions.

E2. Authoritative evidence. Which sources should the copilot treat as authoritative for thresholds and dose–response relationships? For example, should 1000 ppm CO_2 be presented as a ventilation indicator or as a health limit, given the mixed evidence on cognitive effects at typical indoor levels [12, 13]?

E3. Conflicting or context-dependent evidence. How should the copilot communicate uncertainty, for example for the CO_2–cognition link or the choice between the PMV and adaptive comfort models? How should it adapt advice to climate, building type and vulnerable groups such as older people and children?

E4. Licensing of standards. Key standards (ASHRAE, ISO) are copyrighted, and ASHRAE restricts entering its publications into AI tools. Which open sources can substitute for them, and is licensed access for research realistic?

E5. Expert validation. Would you contribute to or review a seed set of Q&A pairs? Which question categories matter most to occupants and to facility managers?

E6. Safety boundaries. Which kinds of advice should the copilot never give, such as medical diagnosis or overriding safety-critical HVAC settings? When should it hand over to a human expert?

E7. Data access. Are there buildings with long-term IEQ and energy sensor data, and ideally a Brick model, that could host the pilot?

Table 1. Candidate knowledge sources identified so far (preliminary; to be confirmed with experts).

Source	Type	What it offers	Limitation for the copilot
Q&A datasets and benchmarks			
BuildingQA [7]	Q&A benchmark	188 questions over Brick graphs of 4 buildings	Metadata queries only; no comfort or health content
AECBench [10]	Q&A benchmark	4,800 expert-reviewed AEC questions, including building codes	Engineering-oriented, not occupant-facing
EnergyGPT benchmark [11]	Q&A benchmark	Domain QA for the energy sector	Broad energy sector; little building or IEQ content
Graph-RAG EE [8]	Q&A validation set	101 Q&A pairs on energy-efficiency guidance	Small; guidance documents only
ASHRAE CHD exam studies (cited in [6])	Exam questions	HVAC design knowledge used to test LLMs	Question sets not released; copyrighted
Evidence sources (content to turn into Q&A)			
ASHRAE Global Thermal Comfort Database II [14]	Open dataset	≈81,846 measurements paired with comfort votes worldwide	Comfort only; no health outcomes
Temperature–office performance database [15]	Open dataset (meta-analysis)	35 studies, 571 performance measures, 20–30 °C	Office work only
Seppänen et al. [16]	Review	Performance peaks near 22 °C; about 9% loss at 30 °C	Older studies; averaged effect
Allen et al. [3]; Du et al. [12]	Experiment; critical review	CO_2/VOC effects on cognition; review of 37 studies with mixed results	Evidence still debated
ASHRAE position on indoor CO_2 [13]	Position document	Expert consensus on CO_2 as an IAQ indicator and open research needs	Not a Q&A resource
WHO Housing & Health [17]; WHO AQG [18]	Guidelines	Minimum indoor temperature in cold seasons (18 °C); PM2.5 guideline values	Population-level, residential focus
CBE occupant survey [19]	Survey analysis	More than 90,000 respondents in about 900 buildings; main sources of dissatisfaction	Raw data not openly available
Cozie / urban comfort dataset [20]	Open dataset	Wearable and environmental data with micro-surveys	Small cohort; short duration

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